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Set 12 :Dataset: Website Traffic

Date Page Views Click	through Rate	
2023	01	01 1500 2.3%
2023	01	02 1600 2.7%
2023	01	03 1400 2.0%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the

chart.

2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart

elements.

3. Build a table to show the website traffic data for each day. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive

exploration of website traffic data.

CODE:

```
import matplotlib.pyplot as plt
```

```
dates = [  
    '2023-01-01',  
    '2023-01-02',  
    '2023-01-03'
```

```
]
```

```
page_views = [1500, 1600, 1400]
```

```
click_rate = [2.3, 2.7, 2.0]
```

```
# 1. Line Chart - Daily Page Views
```

```
plt.figure(figsize=(8, 5))
```

```
plt.plot(  
    dates,  
    page_views,  
    marker='o'  
)
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Page Views")
```

```
plt.title("Daily Page Views")
```

```
plt.show()
```

```
# 2. Bar Chart - Click-through Rate
```

```
plt.figure(figsize=(8, 5))
```

```
plt.bar(dates, click_rate)
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Click-through Rate (%)")
```

```
plt.title("Click-through Rate by Day")
```

```
plt.show()

# 3. Website Traffic Table

print("\nWEBSITE TRAFFIC TABLE")

print("-" * 60)

print("Date\t\tPage Views\tClick-through Rate")

print("-" * 60)


for i in range(3):

    print(

        dates[i],

        "\t",

        page_views[i],

        "\t\t",

        str(click_rate[i]) + "%"

    )print("-" * 60)
```

